

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PHD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Lecture 1

Foundations: Stochastic Processes and ARMA Models

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Topics covered: Stationarity · ACF/PACF · White Noise · MA & AR Processes
ARMA Models · Box–Jenkins Identification · MLE Estimation · Diagnostic Validation ·
Forecasting

Contents

1	Stochastic Processes	2
2	Stationarity	2
2.1	Strict stationarity	2
2.2	Weak (covariance) stationarity	2
3	Autocovariance and Autocorrelation Functions	3
3.1	Autocovariance function	3
3.2	Autocorrelation function (correlogram)	3
3.3	Autocovariance generating function and population spectrum	3
4	Ergodicity	3
5	White Noise	4
5.1	Testing for white noise	4
6	Moving Average Processes	4
6.1	Definition and moments	5
6.2	Invertibility condition	5
6.3	MA(1) worked example	5
6.4	MA(2) worked example	5
7	Autoregressive Processes	6
7.1	Definition	6
7.2	Stationarity condition	6
7.3	Wold's decomposition theorem	6
7.4	Moments and Yule–Walker equations	6
7.5	AR(1) key results	7
8	The Partial Autocorrelation Function	7
9	ARMA Processes	7
9.1	Definition	7
9.2	ARMA(1,1) population spectrum	7
9.3	ARIMA models	8
10	The Box–Jenkins Approach: Identification, Estimation, Validation	8
10.1	Identification	8
10.2	Estimation	8
10.2.1	Exact MLE vs. conditional MLE for AR(p)	8
10.2.2	Newton–Raphson iterative procedure	9
10.2.3	Estimation for MA(q) — backcasting	9
10.3	Validation	9
10.3.1	Recursive least squares	9
10.3.2	CUSUM and CUSUMSQ tests	9

11 Information Criteria for Model Selection	10
12 Forecasting from ARMA Models	10
12.1 Optimal l -step-ahead forecasts	10
12.2 AR(1) forecasts	10
12.3 MA(1) forecasts	11
12.4 ARIMA(0,1,0) with drift (random walk)	11
12.5 Forecasts of transformed variables	11
12.6 Forecast evaluation metrics	11
13 Applied Example: USD/BRL Exchange Rate	11
14 Core References	12

1. Stochastic Processes

A **stochastic process** is a random function indexed by a time set T and a sample space Ω :

$$X : T \times \Omega \longrightarrow \mathbb{R}, \quad (t, \omega) \mapsto X(t, \omega).$$

A specific parametric example is:

$$X(t, \omega) = Y(\omega) \cos[Z(\omega) + t \cdot U(\omega)],$$

where $Y \sim N(0, \sigma_y^2)$, $Z \sim N(0, \sigma_z^2)$, and $U \sim U[-\pi, \pi]$ are mutually independent — a cosine function with a random amplitude, phase, and frequency.

For each time point t we observe only *one realisation* x_t of the random variable X_t . Recovering the full distribution of X_t from a single observation is impossible without further restrictions. Two types of restriction are imposed:

- (i) **Temporal homogeneity** — the statistical structure does not change over time (stationarity);
- (ii) **Limited memory** — distant past observations provide negligible information about the present (ergodicity).

2. Stationarity

2.1. Strict stationarity

Definition 2.1. $\{X_t\}$ is **strictly stationary** if, for every finite collection of indices $\{t_1, \dots, t_n\} \subset T$ and every integer τ ,

$$D(X_{t_1}, \dots, X_{t_n}) = D(X_{t_1+\tau}, \dots, X_{t_n+\tau}).$$

The joint distribution is invariant to time shifts.

2.2. Weak (covariance) stationarity

Definition 2.2. $\{X_t\}$ is **weakly stationary** (or covariance stationary) if:

- (1) $E(X_t) = \mu < \infty, \quad \forall t,$
- (2) $\text{Var}(X_t) = \gamma(0) < \infty, \quad \forall t,$
- (3) $E[(X_{t_1} - \mu)(X_{t_2} - \mu)] = \gamma(t_2 - t_1), \quad \forall t_1, t_2.$

All means and variances are constant, and all covariances depend only on the lag $s = t_2 - t_1$, not on the absolute time. Intuitively, a stationary process *looks the same through any time window* — Valls' window.

3. Autocovariance and Autocorrelation Functions

3.1. Autocovariance function

For a weakly stationary process, the **autocovariance function** (ACF) at lag s is:

$$\gamma(s) = \mathbb{E}[(X_t - \mu)(X_{t-s} - \mu)], \quad s = 0, \pm 1, \pm 2, \dots$$

Proposition 3.1. *The autocovariance function satisfies:*

$$\gamma(0) \geq 0, \tag{1}$$

$$|\gamma(s)| \leq \gamma(0) \quad \forall s \in \mathbb{Z} \quad (\text{Cauchy-Schwarz}), \tag{2}$$

$$\gamma(s) = \gamma(-s) \quad (\text{even function}), \tag{3}$$

$$\gamma(\cdot) \text{ is non-negative definite.} \tag{4}$$

3.2. Autocorrelation function (correlogram)

The **autocorrelation function** normalises the autocovariance by the variance:

$$\rho(s) = \frac{\gamma(s)}{\gamma(0)}, \quad s = 0, 1, 2, \dots$$

Plotting $\rho(s)$ against s gives the *correlogram*.

3.3. Autocovariance generating function and population spectrum

Assuming absolute summability $\sum_{s=-\infty}^{+\infty} |\gamma(s)| < \infty$, the **autocovariance generating function** is:

$$g_X(z) = \sum_{j=-\infty}^{+\infty} \gamma(j) z^j.$$

Setting $z = e^{-i\omega}$ and dividing by 2π yields the **population spectrum**:

$$s_X(\omega) = \frac{1}{2\pi} g_X(e^{-i\omega}) = \frac{1}{2\pi} \left\{ \gamma(0) + 2 \sum_{j=1}^{\infty} \gamma(j) \cos(\omega j) \right\}.$$

Properties: $s_X(\omega)$ is real-valued, non-negative, symmetric around $\omega = 0$, and periodic with period 2π . A key identity links the spectrum to the variance:

$$\sum_{j=-\infty}^{+\infty} \gamma(j) = 2\pi s_X(0) = \gamma(0).$$

4. Ergodicity

Definition 4.1. A stationary process is **ergodic** if $\gamma(s) \rightarrow 0$ as $s \rightarrow \infty$.

Ergodicity is the condition that makes a single time series statistically informative about the underlying data-generating process.

Theorem 4.2 (Ergodic Theorem). *If X_t is strictly stationary and ergodic with $E|X_t| < \infty$, then:*

$$\bar{X}_T = \frac{1}{T} \sum_{t=1}^T X_t \xrightarrow{p} \mu.$$

Theorem 4.3. *If X_t is covariance stationary and ergodic with $E(X_t^2) < \infty$, then:*

$$\hat{\gamma}(s) = \frac{1}{T} \sum_{t=1}^T (X_t - \hat{\mu})(X_{t-s} - \hat{\mu}) \xrightarrow{p} \gamma(s).$$

Moreover, when $\sum_{s=0}^{\infty} |\gamma(s)| < \infty$:

$$\lim_{T \rightarrow \infty} \{T \cdot E(\bar{X}_T - \mu)^2\} = \sum_{j=-\infty}^{+\infty} \gamma(j).$$

5. White Noise

Definition 5.1. $\{X_t\}$ is a *white noise process* if:

$$E(X_t) = \mu, \quad \text{Var}(X_t) = \sigma^2, \quad \gamma(s) = \begin{cases} \sigma^2 & s = 0, \\ 0 & s \neq 0. \end{cases}$$

Equivalently (with $\mu = 0$): $E(X_t | \mathbf{I}_{t-1}) = 0$, so white noise is completely unpredictable from its own past — the benchmark against which all model residuals are tested.

The population spectrum of white noise is flat: $s_X(\omega) = \sigma^2/(2\pi)$ for all ω .

5.1. Testing for white noise

Individual lags. Under homoskedasticity, $\hat{\rho}_s \stackrel{a}{\sim} N(0, 1/T)$, giving a 95% confidence band of $\pm 1.96/\sqrt{T}$.

Ljung–Box joint test.

$$Q^* = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k} \stackrel{a}{\sim} \chi_m^2 \quad \text{under } H_0 : \rho(1) = \dots = \rho(m) = 0.$$

Under heteroskedasticity, use the Bartlett or Newey–West formula for $\text{Var}(\hat{\rho}_s)$.

6. Moving Average Processes

6.1. Definition and moments

A **moving average model of order q** , $\text{MA}(q)$, is:

$$X_t = \mu + u_t + \theta_1 u_{t-1} + \cdots + \theta_q u_{t-q}, \quad u_t \stackrel{\text{iid}}{\sim} (0, \sigma^2).$$

Moments:

$$E(X_t) = \mu, \tag{5}$$

$$\gamma(0) = (1 + \theta_1^2 + \cdots + \theta_q^2) \sigma^2, \tag{6}$$

$$\gamma(s) = \begin{cases} (\theta_s + \theta_{s+1}\theta_1 + \cdots + \theta_q\theta_{q-s})\sigma^2 & 1 \leq s \leq q, \\ 0 & s > q. \end{cases} \tag{7}$$

The ACF of an $\text{MA}(q)$ **cuts off** sharply after lag q . The PACF decays geometrically. Spike count in ACF = q .

6.2. Invertibility condition

An $\text{MA}(q)$ is **invertible** if all roots of $\theta(z) = 0$ lie outside the unit circle. An invertible $\text{MA}(q)$ has an $\text{AR}(\infty)$ representation.

6.3. $\text{MA}(1)$ worked example

Consider $X_t = u_t + \theta u_{t-1}$:

$$\rho(1) = \frac{\theta}{1 + \theta^2}, \quad \rho(s) = 0 \text{ for } s \geq 2.$$

For $\theta_1 = -0.5$: $\rho(1) = -0.4$, $\rho(s) = 0$ for $s \geq 2$.

For $\theta_1 = 0.45$ (EViews simulation): $\rho(1) \approx 0.38$.

Population spectrum: $s_X(\omega) = \frac{\sigma^2}{2\pi} [1 + \theta^2 + 2\theta \cos(\omega)]$.

6.4. $\text{MA}(2)$ worked example

Consider $X_t = u_t + \theta_1 u_{t-1} + \theta_2 u_{t-2}$:

$$\rho(1) = \frac{(\theta_1 + \theta_1\theta_2)}{1 + \theta_1^2 + \theta_2^2}, \tag{8}$$

$$\rho(2) = \frac{\theta_2}{1 + \theta_1^2 + \theta_2^2}, \quad \rho(s) = 0 \text{ for } s \geq 3. \tag{9}$$

For $\theta_1 = -0.5$, $\theta_2 = 0.25$: $\rho(1) \approx -0.476$, $\rho(2) \approx 0.190$.

7. Autoregressive Processes

7.1. Definition

An **autoregressive model of order p** , $AR(p)$, is:

$$X_t = \mu + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + u_t,$$

or compactly using the lag operator $L^i X_t = X_{t-i}$:

$$\phi(L) X_t = \mu + u_t, \quad \phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p.$$

7.2. Stationarity condition

All roots of $\phi(L) = 0$ must lie **outside** the unit circle (equivalently, roots of the characteristic polynomial in z lie *inside* the unit circle).

- $X_t = X_{t-1} + u_t$: characteristic root is 1 (unit root) $\Rightarrow$ non-stationary.
- $X_t = 3X_{t-1} - 2.75X_{t-2} + 0.75X_{t-3} + u_t$: roots are 1, $2/3$, and $2 \Rightarrow$ not all outside unit circle $\Rightarrow$ non-stationary.

7.3. Wold's decomposition theorem

Any stationary process can be expressed as an $MA(\infty)$:

$$X_t = \psi(L) u_t, \quad \psi(L) = \frac{1}{\phi(L)}.$$

This guarantees existence of the variance and autocovariances.

7.4. Moments and Yule–Walker equations

$$E(X_t) = \frac{\mu}{1 - \phi_1 - \cdots - \phi_p}.$$

Autocovariances satisfy the **Yule–Walker equations**:

$$\begin{aligned} \rho(1) &= \phi_1 + \rho(1)\phi_2 + \cdots + \rho(p-1)\phi_p, \\ \rho(2) &= \rho(1)\phi_1 + \phi_2 + \cdots + \rho(p-2)\phi_p, \\ &\vdots \\ \rho(p) &= \rho(p-1)\phi_1 + \cdots + \phi_p. \end{aligned}$$

For a stationary $AR(p)$, the ACF decays *exponentially* (and may oscillate). The PACF cuts off after lag p .

7.5. AR(1) key results

For $X_t = \phi X_{t-1} + u_t$ with $|\phi| < 1$:

$$\text{Var}(X_t) = \frac{\sigma^2}{1 - \phi^2}, \quad (10)$$

$$\gamma(s) = \frac{\phi^s \sigma^2}{1 - \phi^2}, \quad (11)$$

$$\rho(s) = \phi^s, \quad s = 0, 1, 2, \dots \quad (12)$$

Population spectrum: $s_X(\omega) = \frac{\sigma^2}{2\pi[1 + \phi^2 - 2\phi \cos(\omega)]}$.

8. The Partial Autocorrelation Function

The **PACF** at lag k , denoted ϕ_{kk} , measures the correlation between X_t and X_{t-k} after removing the linear effects of all intermediate lags $X_{t-1}, \dots, X_{t-k+1}$. It equals the last coefficient in an OLS regression of X_t on $X_{t-1}, \dots, X_{t-k}$.

Model	ACF	PACF
AR(p)	Decays exponentially / oscillates	Cuts off after lag p
MA(q)	Cuts off after lag q	Decays exponentially / oscillates
ARMA(p, q)	Decays after lag q	Decays after lag p

At lag 1: $\phi_{11} = \rho(1)$. At lag 2: $\phi_{22} = (\rho(2) - \rho(1)^2)/(1 - \rho(1)^2)$.

9. ARMA Processes

9.1. Definition

By combining AR(p) and MA(q) components, we obtain:

$$\phi(L) X_t = \mu + \theta(L) u_t,$$

where $\phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$ and $\theta(L) = 1 + \theta_1 L + \dots + \theta_q L^q$.

Properties: $E(u_t) = 0$, $E(u_t^2) = \sigma^2$, $E(u_t u_s) = 0$ for $t \neq s$.

Mean: $E(X_t) = \mu/(1 - \phi_1 - \dots - \phi_p)$.

For lags beyond q the ACF behaves like a pure AR(p); for lags beyond p the PACF behaves like a pure MA(q).

9.2. ARMA(1,1) population spectrum

For $X_t = \phi_1 X_{t-1} + u_t + \theta_1 u_{t-1}$:

$$s_X(\omega) = \frac{\sigma^2}{2\pi} \cdot \frac{1 + \theta_1^2 + 2\theta_1 \cos(\omega)}{1 + \phi_1^2 - 2\phi_1 \cos(\omega)}.$$

9.3. ARIMA models

An $\text{ARIMA}(p, d, q)$ model is an $\text{ARMA}(p, q)$ applied to the d -th difference of X_t . The integrated process has a characteristic root on the unit circle. Researchers first difference the variable d times until stationarity is achieved, then fit an ARMA to the differenced series.

10. The Box–Jenkins Approach: Identification, Estimation, Validation

10.1. Identification

1. Determine the integration order d (unit root tests — Lecture 2).
2. Use ACF/PACF plots to suggest initial p and q .
3. Apply the *parsimony principle*: prefer the simplest model that adequately describes the data.
4. Select the final specification using information criteria over the entire $\text{ARMA}(p, q)$ grid, keeping the estimation sample constant.

If AR and MA parameters are nearly equal — e.g. an $\text{ARMA}(1,1)$ with $\phi \approx \theta$ — the two polynomials share near-common roots and the ACF/PACF looks like white noise. The model collapses to $\text{ARMA}(0,0)$. Always inspect roots after estimation and apply the parsimony principle.

10.2. Estimation

10.2.1. Exact MLE vs. conditional MLE for $\text{AR}(p)$

Using the prediction-error decomposition:

$$D(X_1, \dots, X_T | \theta) = \prod_{t=p+1}^T D(X_t | \mathbf{I}_{t-1}^{t-p}; \theta) \cdot D(X_1, \dots, X_p | \theta).$$

Exact MLE: includes the unconditional density of the first p observations, $D(X_1, \dots, X_p | \theta) \sim N(\mathbf{0}, \sigma^2 V_p)$. Non-linear in parameters; requires numerical iteration.

Conditional MLE (OLS): treats the first p observations as fixed, maximising the conditional log-likelihood. For $\text{AR}(1)$ this gives a closed form:

$$\hat{\phi} = \frac{\sum_{t=2}^T X_t X_{t-1}}{\sum_{t=2}^T X_{t-1}^2}, \quad \hat{\sigma}^2 = \frac{1}{T} \sum_{t=2}^T (X_t - \hat{\phi} X_{t-1})^2.$$

Differences between exact and conditional MLE are negligible for large T .

Exact MLE log-likelihood for AR(1):

$$\log l(\phi; \sigma^2) = -\frac{T}{2} \ln(2\pi) - \frac{T}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=2}^T (X_t - \phi X_{t-1})^2 + \frac{1}{2} \ln(1 - \phi^2) - \frac{(1 - \phi^2)}{2\sigma^2} X_1^2.$$

10.2.2. Newton–Raphson iterative procedure

The score equations $q(\hat{\theta}) = 0$ are generally non-linear. The Newton–Raphson update is:

$$\theta_{i+1} = \theta_i - H^{-1}(\theta_i) q(\theta_i),$$

where $q(\theta) = \partial \log l / \partial \theta$ and $H(\theta) = \partial^2 \log l / \partial \theta \partial \theta'$. Stopping criteria: convergence in the log-likelihood, score, or parameter values.

10.2.3. Estimation for MA(q) — backcasting

MA parameters have no closed-form estimator. The *backcasting algorithm*:

- (i) Initialise $\hat{u}_0 = 0$ and a starting value θ_0 .
- (ii) Construct $\hat{u}_t(\theta) = X_t - \theta \hat{u}_{t-1}(\theta)$ recursively.
- (iii) Minimise $\sum_{t=1}^T \hat{u}_t^2(\theta)$ with respect to θ .

10.3. Validation

Test two hypotheses after estimation:

- (i) Residuals are Gaussian white noise.
- (ii) Parameters are constant over the sample.

OLS residuals are heteroskedastic and serially correlated by construction and cannot be used directly. Use **recursive (standardised) residuals** instead.

10.3.1. Recursive least squares

The recursive OLS estimator updates as:

$$\hat{\beta}_{t+1} = \hat{\beta}_t + \lambda_{t+1} v_{t+1} (1 + x'_{t+1} \lambda_{t+1})^{-1},$$

where $\lambda_{t+1} = (X_t' X_t)^{-1} x_{t+1}$ and the *one-step-ahead forecast error* (recursive residual) is:

$$v_{t+1} = y_{t+1} - \hat{\beta}_t' x_{t+1}.$$

The standardised recursive residuals $\tilde{v}_{t+1} = v_{t+1} / \sqrt{f_{t+1}}$ are $\stackrel{\text{iid}}{\sim} N(0, \sigma^2)$ under the null of a correctly specified model.

10.3.2. CUSUM and CUSUMSQ tests

CUSUM: $W_t = \sum_{j=k+1}^t \tilde{v}_j / s$. Under parameter constancy $E(W_t) = 0$; divergence from zero signals a break in the mean.

CUSUMSQ: $S_t = \sum_{j=k+1}^t \tilde{v}_j^2 / \sum_{j=k+1}^T \tilde{v}_j^2$. Under constancy $E(S_t) = (t - k) / (T - k)$, a straight line from 0 to 1. Departures signal variance instability.

11. Information Criteria for Model Selection

Let $k = p + q + 1$ be the number of estimated parameters and T the sample size. The three standard criteria are:

$$\begin{aligned} \text{AIC} &= \ln \hat{\sigma}^2 + \frac{2k}{T}, \\ \text{BIC} &= \ln \hat{\sigma}^2 + \frac{k \ln T}{T}, \\ \text{HQIC} &= \ln \hat{\sigma}^2 + \frac{2k \ln(\ln T)}{T}. \end{aligned}$$

- **AIC**: not consistent; tends to select larger models.
- **BIC**: strongly consistent; heavier penalty for extra parameters; preferred in practice.
- **HQIC**: intermediate penalty; also consistent.

A 7×7 grid of ARMA(p, q) models with $p, q \leq 7$ is estimated for the first-differenced log USD/BRL exchange rate. BIC selects an AR(2) as the most parsimonious model with the lowest criterion value (-6.2108 at $(p, q) = (0, 2)$ after accounting for common factors). The two complex roots imply a pseudo-periodic pattern. CUSUM does not reject parameter constancy in the mean, but CUSUMSQ and the normality test reject — evidence of time-varying variance, motivating GARCH (Lecture 5).

12. Forecasting from ARMA Models

12.1. Optimal l -step-ahead forecasts

Define $X_{T+l|T} = E[X_{T+l} | \mathbf{I}_T]$. Forecasts are computed by the rule:

$$X_{T+l|T} = \phi_1 X_{T+l-1|T} + \cdots + \phi_p X_{T+l-p|T} + u_{T+l|T} + \theta_1 u_{T+l-1|T} + \cdots + \theta_q u_{T+l-q|T},$$

where $X_{T+l|T} = X_{T+l}$ for $l \leq 0$ and $u_{T+l|T} = 0$ for $l > 0$, $u_{T+l|T} = u_{T+l}$ for $l \leq 0$.

12.2. AR(1) forecasts

$$X_{T+l|T} = \phi^l X_T, \quad \text{MSE}(X_{T+l|T}) = \sigma^2(1 + \phi^2 + \cdots + \phi^{2(l-1)}).$$

The forecast reverts to the mean as $l \rightarrow \infty$. Under normality:

$$X_{T+l} | \mathbf{I}_T \sim N\left(X_{T+l|T}, \hat{\sigma}^2(1 + \hat{\phi}^2 + \cdots + \hat{\phi}^{2(l-1)})\right).$$

12.3. MA(1) forecasts

$$X_{T+l|T} = \begin{cases} -\theta_1 u_T & l = 1, \\ 0 & l \geq 2. \end{cases}$$

The MA(1) collapses to the unconditional mean after one step.

12.4. ARIMA(0,1,0) with drift (random walk)

$$X_{T+l|T} = X_T + l\mu, \quad \text{Var}(v_{T+l}) = l\sigma^2.$$

The forecast variance grows linearly with the horizon, producing characteristically wide prediction intervals.

12.5. Forecasts of transformed variables

When the model is fitted to $Y_t = \ln X_t$, the optimal forecast of the original series under normality is:

$$\hat{X}_{T+l|T} = \exp(Y_{T+l|T} + \frac{1}{2} \text{Var}(v_{T+l})).$$

12.6. Forecast evaluation metrics

Let $\{X_t\}_{t=T+1}^{T+h}$ be the realised values and $\{X_{t|T}\}$ the forecasts.

Metric	Formula
RMSFE	$\sqrt{\sum_{t=T+1}^{T+h} (X_{t T} - X_t)^2 / h}$
MAE	$\sum_{t=T+1}^{T+h} X_{t T} - X_t / h$
MAPE	$\sum_{t=T+1}^{T+h} X_{t T} - X_t / X_t / h$
Theil's U	$\frac{\sqrt{\sum (X_{t T} - X_t)^2 / h}}{\sqrt{\sum X_{t T}^2 / h} + \sqrt{\sum X_t^2 / h}} \in [0, 1]$

The MSE decomposes into bias proportion, variance proportion, and covariance proportion — useful for diagnosing systematic forecast errors.

13. Applied Example: USD/BRL Exchange Rate

The dataset `bolsa.xls` contains daily data from January 2000 to March 2003 for the Bovespa index, the IBX index, the BRL/USD exchange rate, and the S&P 500.

1. **Identification:** All series in levels exhibit slowly decaying ACFs consistent with stochastic trends. The ADF test fails to reject a unit root in the log exchange rate.
2. **Transformation:** $\Delta \log(\text{Dolar})$ is stationary. Its ACF/PACF suggests $p, q \leq 7$.
3. **Model selection:** BIC over a 7×7 grid selects AR(2) (two complex roots $\Rightarrow$ pseudo-periodic behaviour).

4. **Validation:** Recursive residuals show variance instability near observations 300 and 650. CUSUMSQ and normality tests reject.
5. **Forecasting:** The AR(2) is re-estimated on the sample excluding the last 7 observations. All 7 out-of-sample realisations fall within the 95% prediction intervals, but the bands are wide.

Violation of constant variance in the AR(2) residuals is the canonical motivation for ARCH/GARCH models. The Engle (1982) ARCH test on the squared residuals, and the GARCH(1,1) extension, are covered in Lecture 5.

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These lecture notes draw on `lecture1.tex` (stochastic process foundations and ARMA theory) and `IdentEstValid.tex` (identification, estimation, and validation of ARIMA models) by Prof. Pedro Valls Pereira, EAESP-FGV.